

The signal equation of MRI is given by

$$S_c(t) = \int_V s_c(\mathbf{r}) \rho(\mathbf{r}, t) e^{-i\gamma \int_0^t \Delta B(\mathbf{r}, \tau) d\tau} d\mathbf{r} \quad (1)$$

The difference ΔB from the main magnetic field B_0 can be expressed like

$$\Delta B(\mathbf{r}, t) = \Delta B_0(\mathbf{r}) + \sum_{\beta \in \{x, y, z\}} G_\beta^0(t) r_\beta + \sum_{\substack{|l|, |m| > 0 \\ \beta \in \{x, y, z\}}} G_\beta^{lm}(t) R_l^m(\mathbf{r}) \quad (2)$$

Where $R_l^m(\mathbf{r})$ are the solid harmonics, which can be expressed in terms of the spherical harmonics $Y_l^m(\mathbf{r})$ as

$$R_l^m(\mathbf{r}) = \|\mathbf{r}\|^l Y_l^m(\mathbf{r}). \quad (3)$$

In the ideal case, the higher order terms from the harmonics are zero, and only the linear terms $G_\beta^0(t) r_\beta$ remain. However, in practice there are always some higher order terms present, and the ideal gradient fields prescribed are the actual gradient fields realized. The actual gradient waveforms can be modelled well by a convolution of the ideal gradient waveforms with an impulse response function $\alpha(t)$. This can be expressed for all orders of the gradient fields as

$$G_\beta^{lm}(t) = (\alpha_\beta^{lm} * G_\beta^{lm, \text{ideal}})(t) \quad (4)$$

Or if gradient heating is to be taken into account, the impulse response function can be made time-varying as

$$G_\beta^{lm}(t) = (\alpha_\beta^{lm}(t) * G_\beta^{lm, \text{ideal}})(t). \quad (5)$$

While the magnetization $\rho(\mathbf{r}, t)$ is time dependent, we consider readout to be short enough that the only time dependence is due to T_2 decay. Thus, we can express the magnetization as

$$S_c(t) = \int_V s_C(\mathbf{r}) \rho(\mathbf{r}) e^{-t/T_2(\mathbf{r})} e^{-i\gamma \int_0^t \Delta B(\mathbf{r}, \tau) d\tau} d\mathbf{r} \quad (6)$$

With this we can gather all terms that makes the signal equation deviate from a Fourier transform of the static magnetization in the linear gradient fields and form

$$S_c(t) = \int_V s_C(\mathbf{r}) \rho(\mathbf{r}) e^{\Phi(\mathbf{r}, t)} e^{-i\mathbf{k}(t) \cdot \mathbf{r}} d\mathbf{r}$$

where

$$\Phi(\mathbf{r}, t) = -t/T_2(\mathbf{r}) - i\gamma \int_0^t \Delta B_0(\mathbf{r}) + \sum_{\substack{|l|, |m| > 0 \\ \beta \in \{x, y, z\}}} G_\beta^{lm}(\tau) R_l^m(\mathbf{r}) d\tau$$

$$\mathbf{k}(t) = \gamma \int_0^t (G_x^0(\tau), G_y^0(\tau), G_z^0(\tau)) d\tau$$

Considering a voxel discretization, we can express the operators

$$\begin{aligned} (\mathbf{A}_c)_{kj} &= s_C(\mathbf{r}_j) e^{\Phi(\mathbf{r}_j, t_k)} e^{-i\mathbf{k}(t_k) \cdot \mathbf{r}_j} \\ (\mathbf{A}_c^H)_{ik} &= s_C^*(\mathbf{r}_i) e^{\Phi^*(\mathbf{r}_i, t_k)} e^{i\mathbf{k}(t_k) \cdot \mathbf{r}_i} \\ (\mathbf{A}_c^H \mathbf{A}_c)_{ij} &= \sum_k s_C(\mathbf{r}_j) s_C^*(\mathbf{r}_i) e^{\Phi(\mathbf{r}_j, t_k) + \Phi^*(\mathbf{r}_i, t_k)} e^{-i\mathbf{k}(t_k) \cdot (\mathbf{r}_j - \mathbf{r}_i)} \end{aligned}$$

Now, to get back a Fourier transform, and to enable the Toeplitz property in $\mathbf{A}_c^H \mathbf{A}_c$. We introduce approximations

$$\begin{aligned} \exp(\Phi(\mathbf{r}_j, t_k)) &\approx \sum_{l=0}^L \sigma_{kl}^1 \exp(\Phi(\mathbf{r}_j, t_l)) \\ \exp(\Phi(\mathbf{r}_j, t_k) + \Phi^*(\mathbf{r}_i, t_k)) &\approx \sum_{l=0}^P \sigma_{kl}^2 \exp(\Phi(\mathbf{r}_j, t_l) + \Phi^*(\mathbf{r}_i, t_l)) \end{aligned}$$